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Research paper

Autonomous navigation of a low-cost service robot in a dynamic restaurant environment using enhanced SLAM and GA-optimized fractional PI control

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ABSTRACT

This paper addresses critical challenges in autonomous service robotics for a low-cost restaurant robot equipped with a 3D camera. The proposed framework integrates: (i) a vision-based odometry method using RGB-D data for accurate pose estimation, (ii) an enhanced SLAM system fusing IMU and RGB-D measurements via the Extended Kalman Filter to reduce drift and noise, (iii) a hybrid path planning scheme combining A* for global route optimization with the Dynamic Window Approach for real-time local navigation, and (iv) a Genetic Algorithm–optimized fractional-order PI (GA-FOPI) controller for precise trajectory tracking. Experimental and simulation results show that the GA-FOPI outperforms conventional PI and FOPI controllers in speed, accuracy, and stability. Specifically, the rise time was reduced from 0.35 s to 0.17 s, the settling time from 0.58 s to 0.26 s, and the overshoot nearly eliminated (0.01 %). The integrated system achieves high mapping accuracy, robust navigation, and reliable control in a dynamic restaurant environment. The novelty lies in unifying these techniques into a single cost-effective ROS-based platform, validated through both simulation and real-world deployment, a combination not previously reported in the literature.

1. Introduction

In recent years, the field of service robotics has garnered significant attention and interest among both scholars and commercial entities [1]. The wide range of real and potential applications in this field has motivated researchers and developers to focus on developing autonomous navigation and manoeuvring capabilities for service robots. An autonomous service robot typically requires the following four essential components [2–4]: (i) Perception: The robot utilizes its onboard sensors, such as cameras, lidar, radar, or other sensing devices, to gather data from its surroundings to create the map. It extracts meaningful information from the sensor data, such as identifying objects, detecting obstacles, or recognizing landmarks (ii). Localization: The agent determines its position and orientation in the working space, often called pose estimation. This can be achieved using various techniques, including GPS (Global Positioning System), inertial sensors (accelerometers, gyroscopes), visual odometry, or external tracking systems. Localization is crucial for the robot to know where it is in relation to its target or other objects in the environment. (iii) Cognition and Path Planning: Once the robot has perceived its surroundings and knows its position, it uses its cognition capabilities to make decisions and plan an appropriate path to reach its target. This involves analyzing the collected data, understanding the environment, and choosing the best action. Path planning algorithms consider obstacles, mission objectives, airspace restrictions, and energy efficiency(iv). Motion Control: Once the path is planned, the robot must regulate its motion to follow the desired trajectory. This involves controlling the robot's actuators (such as motors or control surfaces) to adjust its speed and direction.

To enable robots to operate in unknown environments such as restaurants, Simultaneous Localization and Mapping (SLAM) technology is employed. SLAM enables robots to construct a semantic map of the restaurant facilities, capturing spatial layouts and comprehending the surrounding environment. By leveraging SLAM, a robot utilizes its sensors to navigate and explore the unknown environment, simultaneously determining its position and estimating its posture based on observed environmental characteristics. This process involves the robot's continuous movement, data collection from sensors, and the incremental construction of an interior map. The SLAM algorithm allows the robot to perceive its surroundings and gather information from its sensors, such as RGB cameras, depth sensors, or laser scanners. As the robot moves through the restaurant, it analyzes the sensor data to identify features and landmarks within the environment. These features

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may include objects, furniture, walls, or other distinctive elements that help build a comprehensive understanding of the environment [5]. This map can be utilized for various purposes, such as path planning and object recognition. It enables the robot to interact with the environment, navigate the space, and perform tasks efficiently, providing valuable assistance in a restaurant setting. It allows robots to acquire spatial knowledge, understand their surroundings, and incrementally build maps, vital for autonomous navigation, interaction, and efficient task execution within the restaurant setting.

Significant research and development have been implemented in 3D reconstruction, augmented reality, and visual guidance, which rely heavily on self-localization methods [6]. One commonly used method for self-localization is Visual Odometry (VO), which utilizes one or multiple image sensors to estimate the motion and position of a camera or robot in a given environment. Visual odometry calculates the camera's ego motion by analyzing consecutive frames and extracting relevant features, such as corners or critical points, to track their motion. The camera's trajectory can be estimated by comparing the motion of these features over time. VO can be beneficial when other external sensors, such as GPS or IMU, are unavailable or their data is unreliable. In [7], an approach that leverages multi-sensor fusion is proposed, advanced object detection using the YOLO algorithm, and the construction of a 2D topology map to enhance the overall capabilities of the robotics system. Using an IMU improves motion estimation and sensor data fusion, while YOLO-based semantic information extraction and topology mapping contribute to a more comprehensive understanding of the environment.

On the other hand, Simultaneous Localization and Mapping (SLAM) takes the concept of Visual Odometry further by simultaneously constructing a map of the environment while estimating the camera or robot's pose within that map [8,9]. SLAM algorithms consistently represent the environment by combining the camera's motion estimation with the 3D features extracted from the images [10]. SLAM algorithms typically operate in real-time, incrementally updating the map and estimating the camera's pose as new frames are captured [11]. By fusing information from multiple frames and features, SLAM algorithms can handle loop closures (revisiting previously seen areas) and optimize the map and camera poses for increased accuracy. Visual Odometry and SLAM have become crucial technologies in various applications, including robotics, augmented reality, autonomous vehicles, and more. They provide the means for machines to understand their surroundings and navigate effectively in real-world environments [12]. Recently, several researchers have employed hybrid and meta-heuristic optimization to enhance the performance of fractional-order PID controllers in nonlinear dynamic systems [13,14]. Similar optimization-based FOPID approaches have also been successfully applied to nonlinear and dynamic robotic systems, confirming their robustness and adaptability when tuned using meta-heuristic algorithms [15–17]. Hybrid optimization has also been applied to improve UAV trajectory planning, where the combination of fractional-order control and meta-heuristics achieved smoother and shorter flight paths [18]. Multi-criteria decision--making frameworks such as CRITIC and TOPSIS have been incorporated with hybrid algorithms to assess stability, precision, and robustness of UAV control systems [19]. A deep-learning real-time visual SLAM system based on a multi-task feature extraction network and self-supervised feature points is proposed in [20]. Designing a simplified Convolutional Neural Network (CNN) for feature point detection and descriptor extraction can enhance the accuracy and stability of a visual SLAM system. The key idea behind using a CNN is to leverage its ability to learn discriminative features directly from raw image data, thereby improving the robustness and adaptability of the feature extraction process. An approach based on the ORB-SLAM2 system with stereo observations using deep learning models and traditional techniques is investigated in [21]. The approach combines deep learning-based semantic analysis, depth estimation, and traditional SLAM techniques to enhance the accuracy and stability of the visual SLAM system. The multiple viewpoints and careful feature selection help mitigate challenges posed by moving objects and occlusions. Incorporating semantic information and depth cues improves the system's understanding of the scene, leading to more reliable navigation. The creation of multiple dense 3D maps in an incremental manner ensures that the system remains adaptable to changes in the environment. Integrating the various components and rigorous testing under real-world conditions would be necessary to implement this approach effectively. In [22], evaluating and enhancing Visual SLAM solutions within the context of the ROS (Robot Operating System) middleware is developed. First, it addresses the real-world constraints of running Visual SLAM algorithms on embedded platforms like the NVIDIA Jetson TX2, and second, it demonstrates the benefits of utilizing the embedded GPU for image processing tasks. These contributions are relevant for the robotics community, as they offer insights into making Visual SLAM algorithms more practical and efficient for real-world applications, particularly in scenarios where computational resources are limited.

To adjust the robot's speed, direction, and altitude, various control techniques can be employed depending on the specific application and robot type [23]. Fractional order controllers have gained significant attention in recent years and are being explored by scientists and researchers to achieve robust performance in various systems [24]. In traditional control systems, integer-order controllers assume that the system dynamics are purely integer-order (proportional-integral-derivative PID), i.e., the orders of differentiation and integration are restricted to integer values. However, many real-world systems exhibit non-integer or fractional-order behavior, meaning their dynamics are better described using fractional calculus [25]. In [26], a model-free implementation and output feedback control for robust tracking of robotic manipulators is proposed. In the approach, the PID error manifold is synthesized to improve the tracking capabilities of the closed-loop system. Additionally, a dynamic controller is built upon a Takagi-Sugeno inference system to ensure that the PID extended error evolves close to the origin. This inference system is based on the PID extended error and its fractional order (FO) integral. In [27], the authors recommend evaluating the performance of both classical PID and fractional PID controllers in the target application, considering factors such as stability, disturbance rejection, robustness, and tracking accuracy to determine the most suitable control strategy.

Service robots are inherently complex systems that require precise control and coordination of various subsystems to operate effectively. Integrating image-based deep learning, ROS (Robot Operating System), and optimal control significantly enhances their navigation and operational capabilities. This integration enables the estimation of the robot's pose, obstacle detection, path generation, and parametric optimization of its optimal controller. However, current service robots that are affordable are still not able to fully meet these requirements. The current study proposes a real-time implementation for a low-cost, designed service robot autonomous navigation to trigger its identification, 3D vision-based SLAM algorithm, detect obstacles, generate the appropriate trajectory, and perform an optimal Fractional PI controller. This study aims to demonstrate that autonomous navigation is feasible on low-cost-designed service robots. The paper presents three key contributions:

- Development of Odometry Method using 3D Vision in Unknown Environment: The paper introduces a localization method that utilizes 3D vision data from an RGB-D camera to estimate the position and orientation of a robot in an unknown environment. This odometry method addresses the challenge of dealing with data provided by a 3D camera and demonstrates how it can be used to obtain accurate localization states
- Integrating IMU and RGB-D data using EKF enhances SLAM to capitalize on the complementary strengths of the sensors, reducing drift error and diminishing noise, resulting in a more reliable and robust SLAM system.

- Integrating A* for global path planning and the Dynamic Window Approach (DWA) for local path planning for navigation in unknown environments. This approach leverages A* to find an optimal path from the start to the goal, considering static obstacles, and DWA to navigate dynamically around obstacles that may not have been known initially.
- Design and Implementation of Optimal FOPI Controller based on Genetic Algorithm (GA): The paper introduces the design and implementation of an optimal fractional order proportional-integral (FOPI) controller using a Genetic Algorithm (GA). The controller is specifically tailored for tracking reference trajectories in a service system. The simulations conducted in the paper compare the proposed GA-FOPI controller with conventional PID controllers and other existing FOPID controllers. The results demonstrate the effectiveness and superiority of the proposed GA-FOPI controller over other controllers in terms of stair response analysis and performance evaluation.

The main novelties of this study are: A robust RGB-D odometry pipeline combining SURF, NNDR, PnP, and RANSAC for accurate pose estimation in unknown indoor spaces; An improved SLAM system by fusing IMU and RGB-D data through EKF to reduce drift and noise, enhancing mapping accuracy on a ROS platform; A hybrid path planning framework integrating A* for optimal global route planning with DWA for real-time dynamic obstacle avoidance; A GA-optimized fractional-order PI controller tailored for trajectory tracking, outperforming conventional PI and FOPI controllers in simulations and physical trials. This integrated design, applied to a low-cost service robot and validated in real-world restaurant settings, has not been reported previously. To the best of our knowledge, no prior study has implemented and validated this full integration in a dynamic restaurant scenario.

In summary, although several previous studies have investigated FOPID or hybrid optimization controllers for mobile robots, most rely on high-end hardware or focus on single-module improvements. The main novelty of this study lies in integrating multiple established methods—odometry, SLAM, hybrid path planning, and GA-FOPI control—into a unified, low-cost framework validated in real-world restaurant scenarios. This work highlights the feasibility of achieving reliable navigation using affordable sensors and open-source software, addressing the challenge of balancing performance, cost, and computational efficiency.

Section II reviews related works and system overview. Section III details the proposed methodology, including odometry, SLAM, hybrid path planning, and control design. Section IV presents experimental validation and analysis, followed by discussions in Section V and engineering applications in Section VI. Section VII concludes the paper.

2. Mobile robot design

The designed service robot prototype is in Fig. 1. The hardware includes primary modules: Mechanical design; LCD to display and communicate with customers; an Intel *Re-* alsense RGB-D camera used for SLAM and detect obstacles; electronics design; software design so that the robot can build a map and transport food from the counter area to the tables at the request of diners, interacting directly with customers through the GUI. The robot is designed compactly for restaurant space: $50 \times 40 \times 1000$ mm, as shown in Fig. 1. The robot has a maximum speed of 0.5 m/s without a load and a maximum speed of 0.3 m/s with a load.

Fig. 2 presents all modules that are implemented for our robot service system. The robot works based on the primary sensors, which are IMU sensors, 3D cameras, ultrasound sensors, and an encoder to orient robots, determine the correct direction, build maps, and detect obstacles and places of altitude difference. This sensor will send signals to the Raspberry Pi 4 microcomputer for processing and giving out the proper navigation methods. Wireless communication is crucial for maintaining remote control of a robot and enabling efficient robot operation. The robot is controlled via a master station on the ROS platform.

3. Odometry, slam, path planning algorithm, and obstacle detection

This section presents the methodology of the proposed system, describing the perception, mapping, path planning, and control components that enable autonomous navigation. Each module of odometry,



Fig. 1. Designed Robot Equipped 3D camera and LCD: Left - design 3D robot; Right: Real designed robot.

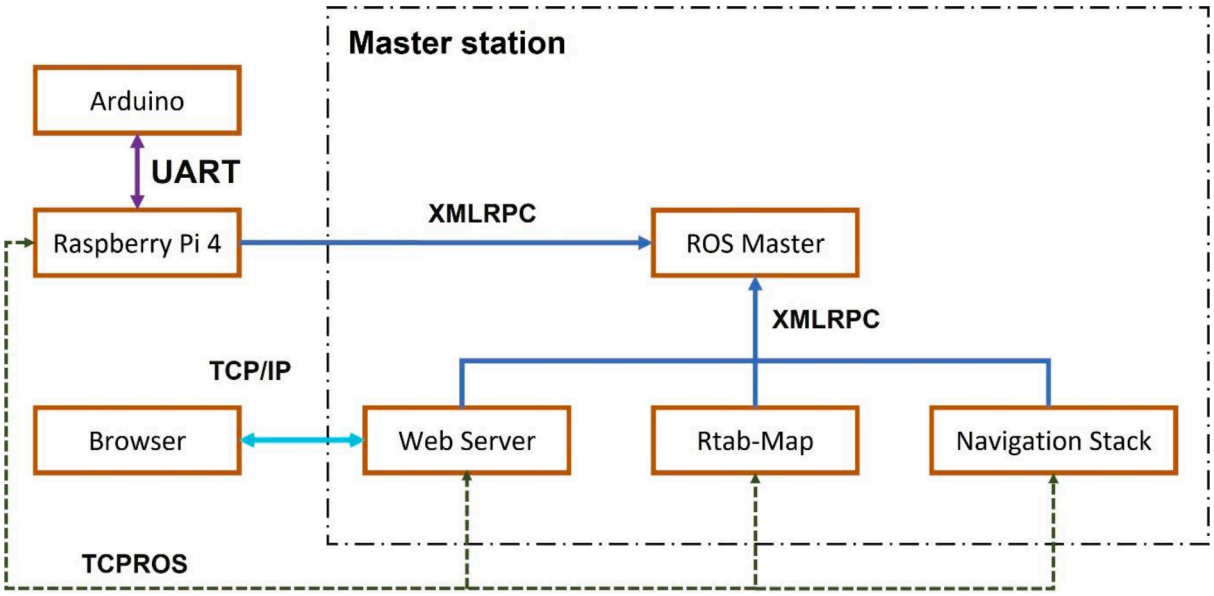


Fig. 2. Software and hardware structure of our service robot.

SLAM, obstacle detection, and hybrid path planning is detailed to explain how they interact within the integrated framework.

3.1. Odometry, SLAM-based service robot

This subsection describes the odometry process, which estimates the robot's motion using RGB-D visual features and IMU data to obtain reliable pose information.

In our study, RTAB-Map is directly used for both odometry and SLAM as it is a powerful software package designed for 3D mapping and navigation using RGB-D cameras. Its versatility, real-time processing capabilities, and robust features make it suitable for a wide range of autonomous robot navigation. Odometry in the context of RGB-D cameras with RTAB-Map involves estimating the motion of the camera over time by analyzing the changes in the RGB (color) and depth images. The odometry process is presented in Fig. 3. The SURF algorithm is used to extract the feature in the image using Fast Hessian [28]. Then, the feature matching is implemented using Nearest Neighbor Distance Ratio (NNDR) by considering the distance ratio to the first nearest neighbor and the distance to the second nearest neighbor [29]. The methodology combined the Perspective-n-Point (PnP) problem and the Random Sample Consensus (RANSAC) algorithm used to deal with noisy or incomplete data, where some correspondences between 3D points and their 2D projections might be incorrect due to factors like image noise, occlusions, or mismatches. This is used to calculate the transformation of the current frame according to the features in the feature map. However,

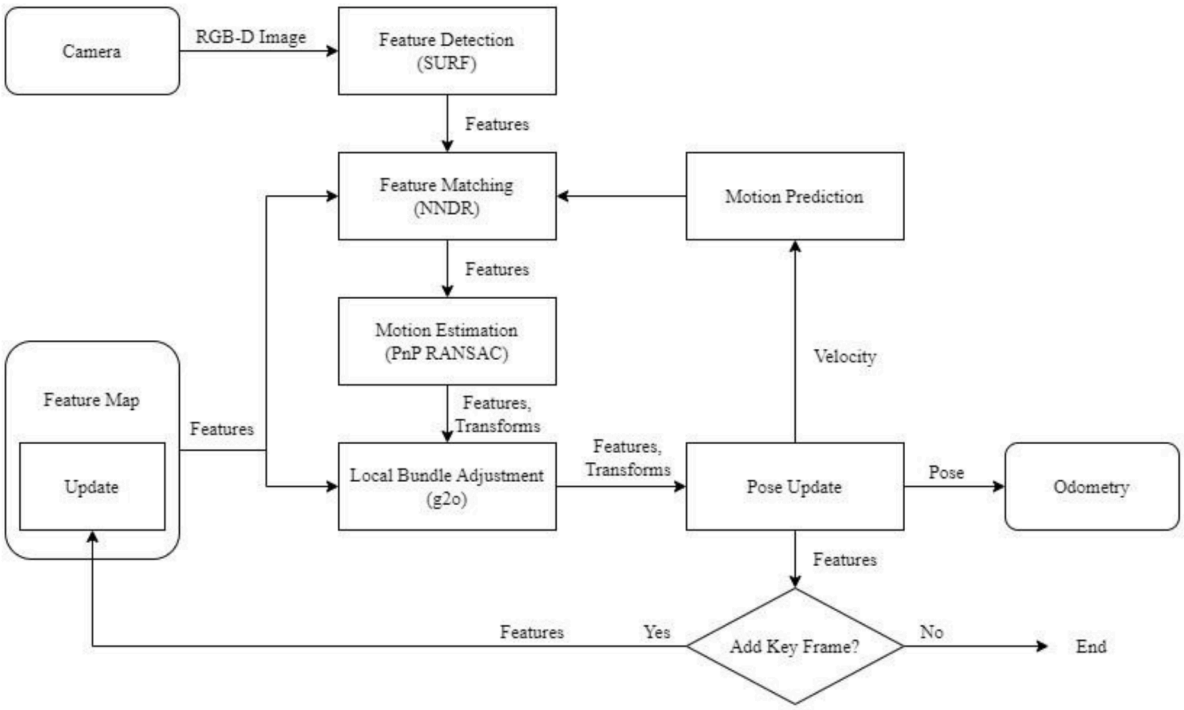


Fig. 3. Vision based odometry methodology.

the odometry performance may degrade over long trajectories or in feature-sparse areas, which remains a limitation of the current configuration.

In this study, the RGB-D odometry module provided by RTAB-Map was employed in its standard configuration using SURF for feature extraction, NNDR for feature matching, and PnP–RANSAC for pose estimation. No algorithmic modification was made to these components. Instead, the contribution of this work lies in the system-level integration of this standard odometry process with IMU-based sensor fusion via EKF, hybrid A–DWA path planning*, and GA-FOPI motion control on a low-cost mobile robot platform. This integration demonstrates that reliable 3D perception, mapping, and control can be achieved without proprietary or high-end hardware, emphasizing practical applicability over algorithmic novelty.

To simplify the problem, we now consider the camera's coordinate system to coincide with the robot's coordinate system. Since the camera is mounted on the robot, the coordinate system can easily represent the degree of the robot according to the camera's coordinate system. The position of two cameras at two consecutive times $k-1$ and k are related to each other by the transformation as in the equation.

$$T_{k,k-1} = \begin{bmatrix} R_{k,k-1} & t_{k,k-1} \\ 0 & 1 \end{bmatrix} \quad (1)$$

where $R_{k,k-1}$ is a rotation matrix; $t_{k,k-1}$ is a translation vector.

The proposed methodology follows a systematic workflow integrating perception, mapping, planning, and control stages. First, RGB-D and IMU data are fused to perform vision-based odometry for accurate pose estimation. The estimated poses are then fed into an enhanced SLAM module, where the Extended Kalman Filter (EKF) minimizes drift and sensor noise during map construction. Once the environment map is generated, a hybrid path planning module determines the robot's movement: the A* algorithm computes the optimal global path, while the Dynamic Window Approach (DWA) refines real-time local obstacle avoidance. The final motion control is handled by the GA-optimized fractional-order PI (GA-FOPI) controller, which regulates wheel velocities to achieve smooth trajectory tracking. The overall data flow, equations, and interactions among these modules are detailed in this section and supported by the quantitative results presented in Tables 1 and 2 and Figs. 10–13.

To obtain a 3D-reconstructed mapped point cloud for our restaurant, we developed an RGB-D SLAM that combines an IMU (Inertial Measurement Unit) and an RGB-D camera. Fusing data from both equipment allows for more accurate and robust SLAM, especially in challenging environments. The EKF combines high-frequency IMU data with lower-frequency but richer information from the RGB-D camera. The IMU helps to predict the camera's movement, while the RGB-D camera provides corrections to this prediction, reducing drift and improving accuracy. The process involves pre-processing, initial pose estimation, sensor fusion, optimization, and loop closure detection. The IMU provides high-frequency motion updates, while the RGB-D camera provides accurate environmental measurements, with the EKF combining these to maintain an accurate and consistent state estimate. This reduction in effort and time enhances construction quality and safety by providing more efficient data acquisition and processing. The proposed algorithm provides a high-resolution registered RGB-mapped point cloud. The SLAM structure is presented in Fig. 4.

Table 1
Summary of the optimal FOPI robustness analysis.

Parameters	PI simulation	PI experiment	FOPI simulation	FOPI experiment
K_p	20	48	20	48
K_i	15	30	15	30
λ	1	0.9	1	0.9
$T_r(s)$	0.35	0.4	0.17	0.2
$T_s(s)$	0.58	0.65	0.26	0.3
OS (%)	0.18	0.1	0.09	0.01

Table 2
Statistical summary of controller performance under various test cases ($N = 10$).

Setpoint (m/s), Noise Range	Controller	ITAE (mean ± SD)	Overshoot (%)	Settling Time (s)
4, (–0.005, 0.005)	PID	3.952 ± 0.035	0.92 ± 0.01	1.41 ± 0.01
	FOPID	3.325 ± 0.031	0.13 ± 0.00	0.99 ± 0.01
	GA-FOPID	**3.186 ± 0.032**	**0.57 ± 0.03**	**0.54 ± 0.00**
10, (−0.01, 0.01)	PID	8.363 ± 0.047	0.90 ± 0.01	1.41 ± 0.00
	FOPID	6.788 ± 0.047	0.10 ± 0.00	0.99 ± 0.02
	GA-FOPID	**6.436 ± 0.046**	**0.55 ± 0.04**	**0.54 ± 0.00**
2, (−0.008, 0.008)	PID	5.375 ± 0.036	1.18 ± 0.02	1.42 ± 0.02
	FOPID	5.077 ± 0.036	0.40 ± 0.00	1.10 ± 0.05
	GA-FOPID	**5.013 ± 0.036**	**0.78 ± 0.06**	**0.54 ± 0.01**
15,(−0.003, 0.003)	PID	5.579 ± 0.025	0.83 ± 0.00	1.41 ± 0.00
	FOPID	2.745 ± 0.023	0.02 ± 0.00	0.98 ± 0.00
	GA-FOPID	**2.341 ± 0.024**	**0.52 ± 0.01**	**0.54 ± 0.00**
1, (−0.001, 0.001)	PID	0.834 ± 0.007	0.90 ± 0.01	1.41 ± 0.00
	FOPID	0.676 ± 0.007	0.10 ± 0.00	0.98 ± 0.01
	GA-FOPID	**0.641 ± 0.007**	**0.55 ± 0.04**	**0.54 ± 0.00**

3.2. Obstacle detection and path planning algorithm

This subsection presents the SLAM module, which fuses RGB-D and IMU data via the Extended Kalman Filter to achieve accurate real-time mapping and localization.

a) *Obstacle detection:* Object detection was implemented using YOLOv3 as an integrated perception module to identify obstacles in the restaurant environment. The pretrained YOLOv3 model (trained on the COCO dataset) was directly employed to detect key object classes, including person, chair, table, and tray. No additional fine-tuning or dataset-specific retraining was performed, as the pre-trained model provided sufficient accuracy and robustness for safe indoor navigation on our low-cost onboard hardware. The perception module was designed to operate efficiently within the integrated SLAM–planning–control framework. In the restaurant scenario, several objects needed to be detected and avoided, such as tables, chairs, and humans. The proposed obstacle detection approach is based on the YOLOv3 algorithm [30]. Images captured from the 3D camera are continuously fed into the YOLOv3 network, where detected objects are identified and marked with bounding boxes in the corresponding frames. The detection results are illustrated in Fig. 5. Quantitative evaluation metrics such as precision, recall, and F1-score were not computed in this study, as YOLOv3 was used as an off-the-shelf perception module to support navigation rather than as a novel detection method.

b) *Path planning algorithm:* Path planning is designing a collision-free path in a working environment with obstacles. A service robot path is a set of configurations $q: = \{q_0, q_1, \ldots, q_{goal}\} \in R^n$ of the agent that connects the starting position q_0 and the final position q_{goal}. The q_{goal}.

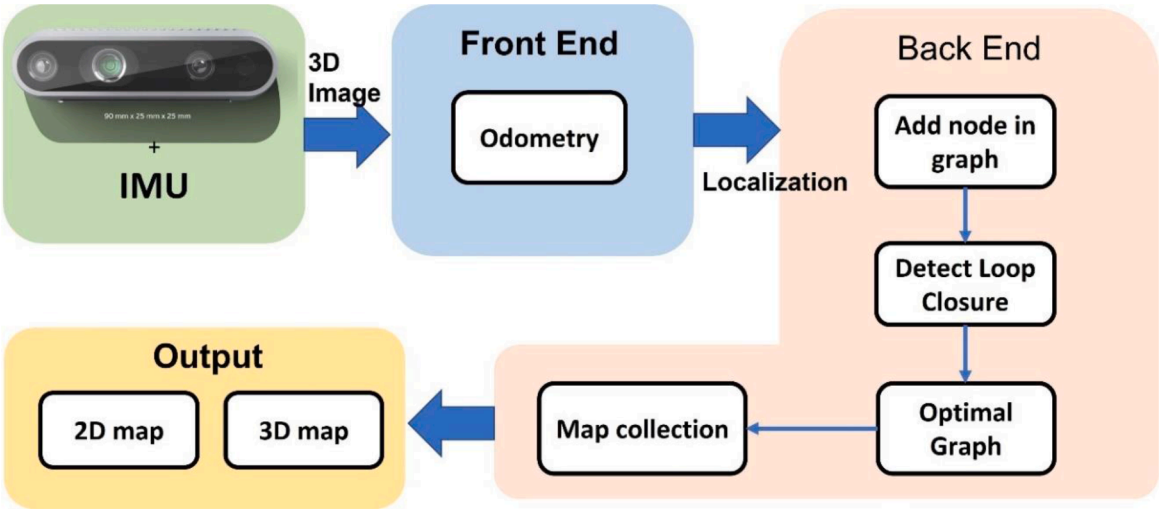


Fig. 4. The SLAM structure.

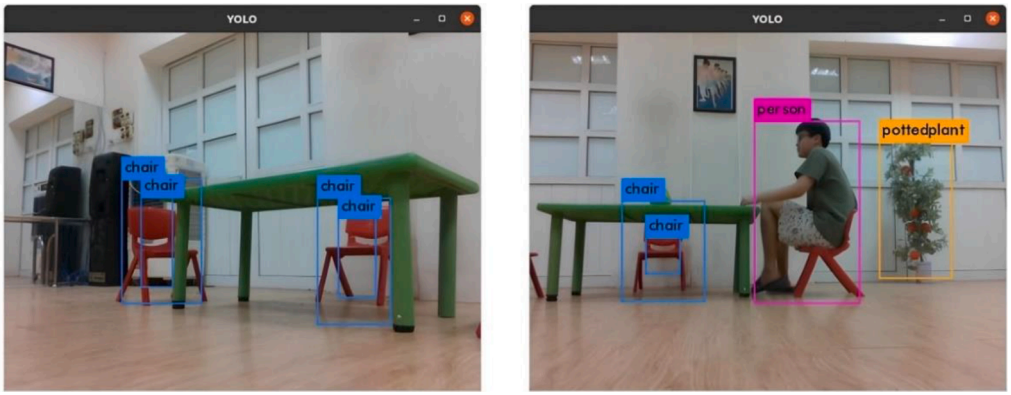


Fig. 5. Detection Results using deep learning techniques of the Service Robot.

In our application, we propose the path planning algorithm on ROS for the robot based on the A* algorithm to find the globally optimal path and DWA (Dynamic window approach) to find the optimal local path, as presented in Fig. 6. The objective function used to calculate the optimal path is defined as follows:

$$G(v,\ \omega) = \alpha heading(v,\ \omega) + \beta dist(v,\ \omega) + \gamma vel(v,\ \omega) \quad (2)$$

where:

$v,\ \omega$ are the linear velocity and angular velocity of the robot.

$dist(v,\ \omega)$ is the distance to the obstacles.

$heading(v,\ \omega)$ is the function calculate the direction of the robot and

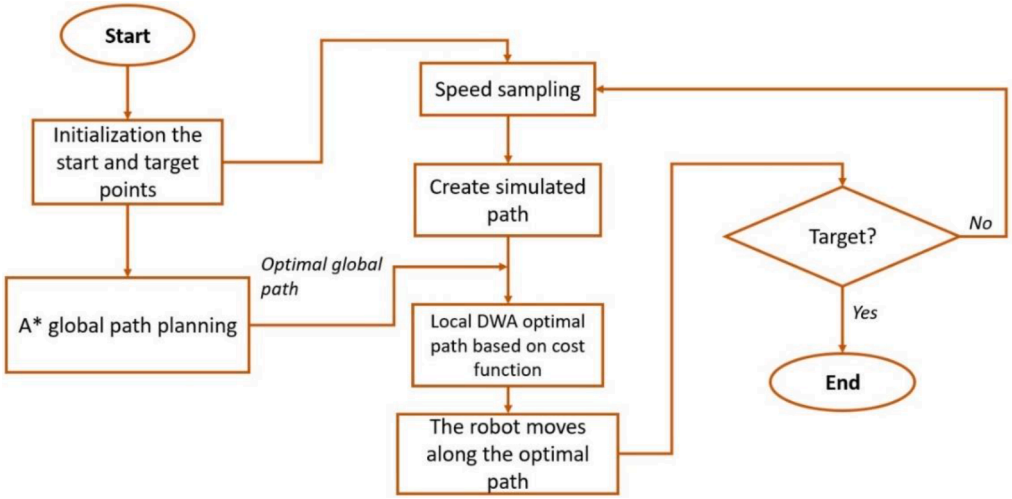


Fig. 6. ROS-based path planning algorithm.

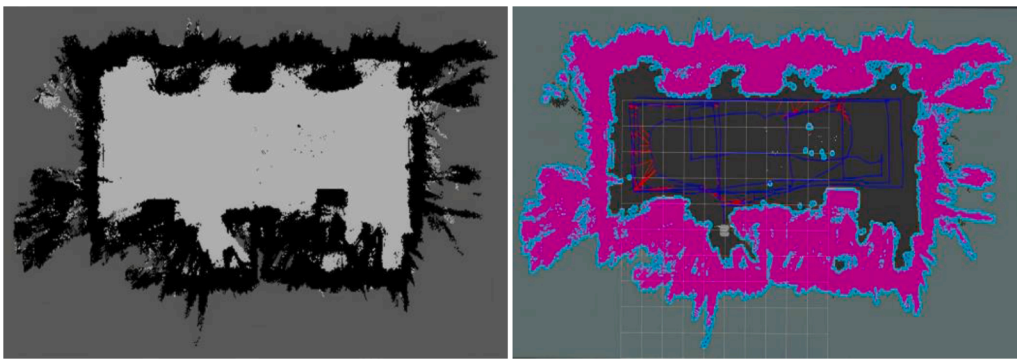


Fig. 7. 3D Trajectory using point cloud.

the target.

$vel(v,\ \omega)$ is used to evaluate the robot's progress on the robot trajectories.

$\alpha,\ \beta,\ \gamma$ are weight coefficients.

All components of the objective function are normalized to the interval $[0, 1]$. The weight coefficients ($\alpha = 0.8$, $\beta = 0.1$, $\gamma = 0.1$) in the DWA evaluation function were determined empirically through multiple tuning trials in a restaurant-like indoor environment. The coefficient $\alpha = 0.8$ emphasizes heading alignment to ensure goal-directed motion efficiency, $\beta = 0.1$ accounts for obstacle clearance considering tables, chairs, and human movement, while $\gamma = 0.1$ maintains adequate forward velocity for smooth navigation. This combination was found to yield stable and safe trajectories on the low-cost platform.

It should be noted that all three components of the objective function $G(v,\ \omega)$, including target direction, distance total, and velocity, are needed. Only by maximizing the safe distance and speed will the robot always move into the safe zone, but not to the target location. Combining all three components allows the robot to avoid collisions as quickly as possible, progressing toward the goal. Fig. 7 illustrates the reconstructed 3D trajectory obtained from the RGB-D SLAM process, showing that the robot maintains continuous pose estimation and a smooth navigation path within the restaurant environment.

4. Robot identification and optimal control design

This section introduces the identification of the robot dynamics and the design of the optimal fractional-order PI controller tuned by a Genetic Algorithm. It explains how the controller parameters are optimized to improve speed, accuracy, and robustness while maintaining computational efficiency on a low-cost platform.

The PID (Proportional-Integral-Derivative) controller is a cornerstone of control systems and is widely used for its simplicity and effectiveness in various applications. However, the demand for better control performance and increased robustness in modern control systems has led to several modifications and enhancements of the standard PID controller. One significant advancement is the generalization to fractional-order PID controllers, commonly referred to as FOPID [31, 32]. In [33], a Fractional Order Proportional- Derivative (FOPD) controller for multiple Unmanned Aerial Vehicles (UAVs) in an emergency health case is developed. The controller needs to ensure that the UAVs operate with high precision and reliability, especially in sensitive scenarios such as medical emergencies. The synthesis of a rotational speed closed-loop control system based on a fractional-order proportional-integral (FOPI) controller is developed for a Poorly Damped fractional-order process with Time Delay [32]. In this section, we develop an optimal fractional PI control for a service robot. In this section, we develop an optimal fractional PI control for a service robot. In our proposed method, the hyperparameters (Kp, Ki, λ) of the FOPI are archived by a GA-based approach to minimize rise time, settling time, and overshoot. First, we identify the dynamics of the system to be controlled by using the methodology presented in [3]. The transfer function of the system is obtained as:

$$G(s) = \frac{0,55}{S^2 + 31.17s + 165.8} \tag{3}$$

This involves obtaining a mathematical model or a transfer function that describes the system's behavior. In general, the FOPID is defined as in Eq. (4)

$$C_{FO-PID}(s) = k_p + k_i s^{-\lambda} + k_d s^{\mu} \tag{4}$$

The effectiveness of the λ and μ are presented in Fig. 8. In this study, the transfer function of the fraction order PI $(FO-PI)$ is designed as follows:

$$H_{FO-PI}(s) = k_p\left(1 + \frac{k_i}{s^{\mu}}\right) \tag{5}$$

where: μ is the fractional order and k_p, k_i are the proportional, integrative gains, respectively.

In fractional PI control, the integral and proportional actions are modified by introducing fractional order operators, such as fractional integrators and differentiators. These operators allow for non-integer orders μ between 0 and 1, enabling more accurate modelling and control of systems with fractional dynamics. It is worth noting that the design and implementation of fractional PI control can be more complex compared to traditional PI control due to the additional fractional order parameters. However, it offers advantages in dealing with systems exhibiting fractional dynamics, such as non-linearity, memory effects, or long-range dependencies.

The proposed approach structure of this study is presented in Fig. 9. The objective function is designed as follows.

$$J_{ITAE} = \int_0^{t_f} t|e(t)|dt \tag{6}$$

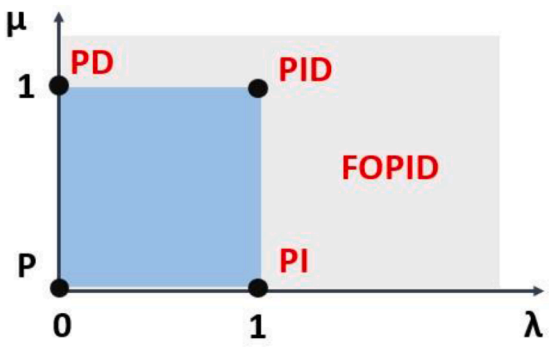


Fig. 8. Effect of coefficients λ and μ on the FOPID controller.

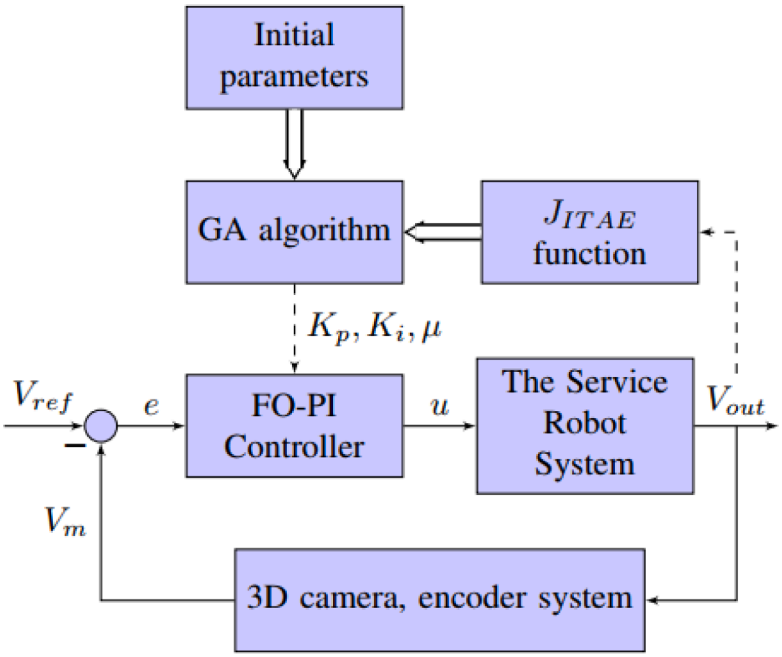


Fig. 9. The proposed optimal GA-based FO-PI controller approach for a service robot.

where: $e(t) = V_{ref}(t) - V_m(t)$ and t_f is the final simulation time. The GA searches for the FOPI parameters (K_p, K_i, λ) that minimize this criterion, leading to an optimal balance between speed, smoothness, and steady-state accuracy.

Having the service robot model (section IV), the controller parameters are obtained by applying the proposed GA-based FO-PI controller. In this study, the GA parameters were empirically selected as a population size of 30, a crossover fraction of 0.8, and a mutation rate of 0.05. These values were determined after multiple tuning trials to balance convergence reliability, solution quality, and computational feasibility on the low-cost platform. The optimization used the Integral of Time-weighted Absolute Error (ITAE) as the performance criterion because ITAE penalizes prolonged steady-state errors, leading to smoother and faster responses suitable for trajectory tracking in service-robot applications. Although other criteria, such as ISE and IAE, were also considered, ITAE provided the most stable and consistent results in both simulation and real-world experiments. The parameters of the GA algorithm are provided as follows:

Fitness scaling function: fitscalingrank
Mutation function: mutationuniform
Population size: 25
Max generations: 250
Constraint tolerance: 0.001
Crossover function: crossoversinglepoint
Crossover fraction: 0.8

The results of the GA algorithm are presented in Fig. 10. FOPI controllers' optimal parameters are obtained through simulations. The optimal *Kp*, *Ki*, λ parameter sets obtained by the proposed GA-based FOPI with rise time, settling time, and overshoot the compared with PI simulation/ PI experiment/ FOPI simulation in Table I.

To evaluate the consistency and statistical significance of the results, each controller was tested ten times under different setpoints and noise levels. The performance indices were calculated as mean ± standard deviation (SD) for the ITAE, overshoot, and settling time metrics. Table 2 summarizes the statistical outcomes across all test cases. The statistical results indicate that the GA-FOPID controller consistently achieved the lowest ITAE and settling time across all test cases, with small standard deviations. This demonstrates not only improved mean performance, but also higher consistency and robustness compared with the baseline controllers. The small variability confirms that the observed gains are statistically meaningful rather than random fluctuations.

5. Experimental validation and analysis

In this section, we provide experimental evidence to prove the feasibility and effectiveness of our proposed approach practically. Robot Operating System (ROS) is used for our service robot, developed at the Hanoi University of Science and Technology. The experimental setup involves a mobile robot with two critical sensors: an Rplidar-A1 sensor and an Intel RealSense D435 sensor. The lidar sensor generates an outline map of the robot's working environment using clockwise rotation, with a range of up to 12 m The Intel RealSense D435 sensor

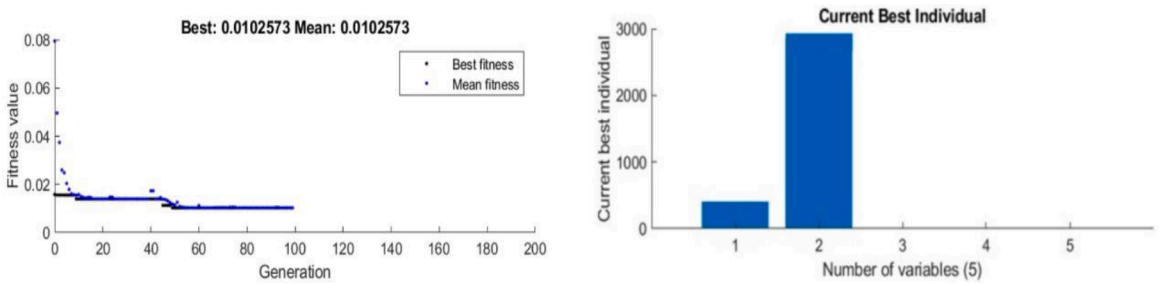


Fig. 10. Minimum and average value of the objective function (Left) and the best results are selected (Right).

provides real-time 3D information with precision for obstacle avoidance, as mentioned in subsection 3.1. The 3D camera is also used to provide the 3D information of the restaurant. The test case takes place in an indoor environment, specifically a tiny restaurant that contains furniture like tables and chairs, and a counter table. The indoor environment spans approximately 30 square meters.

The experimental results align with the simulated evidence presented in Section III of this work. This implies that the outcomes achieved through simulation are also observable and valid in real-world scenarios. We have tested and verified various components and algorithms as part of our approach, including odometry, SLAM (Simultaneous Localization and Mapping), path planning algorithms, and control mechanisms. The experimental results are consistent with the comparison results presented in Section IV of this work. The alignment between simulation, experimental results, and comparisons helps validate the reliability of the proposed approach for controlling the service robot's behaviour in a real-world environment.

Fig. 11 compares different controllers for a service robot's wheel velocity using a performance metric - Integral of Time- weighted Absolute Error (ITAE) criterion. Blue solid curves represent the GA-FOPI controller, while the PI controller is represented by red dot curves in the figure. The performance of the controllers is evaluated using the JIT AE function, a criterion used to assess control system performance based on the integral of the time-weighted absolute error. The results suggest that the GA-FOPI controller outperforms the PI controller in performance. The GA-FOPI controller is noted to exhibit better performance,

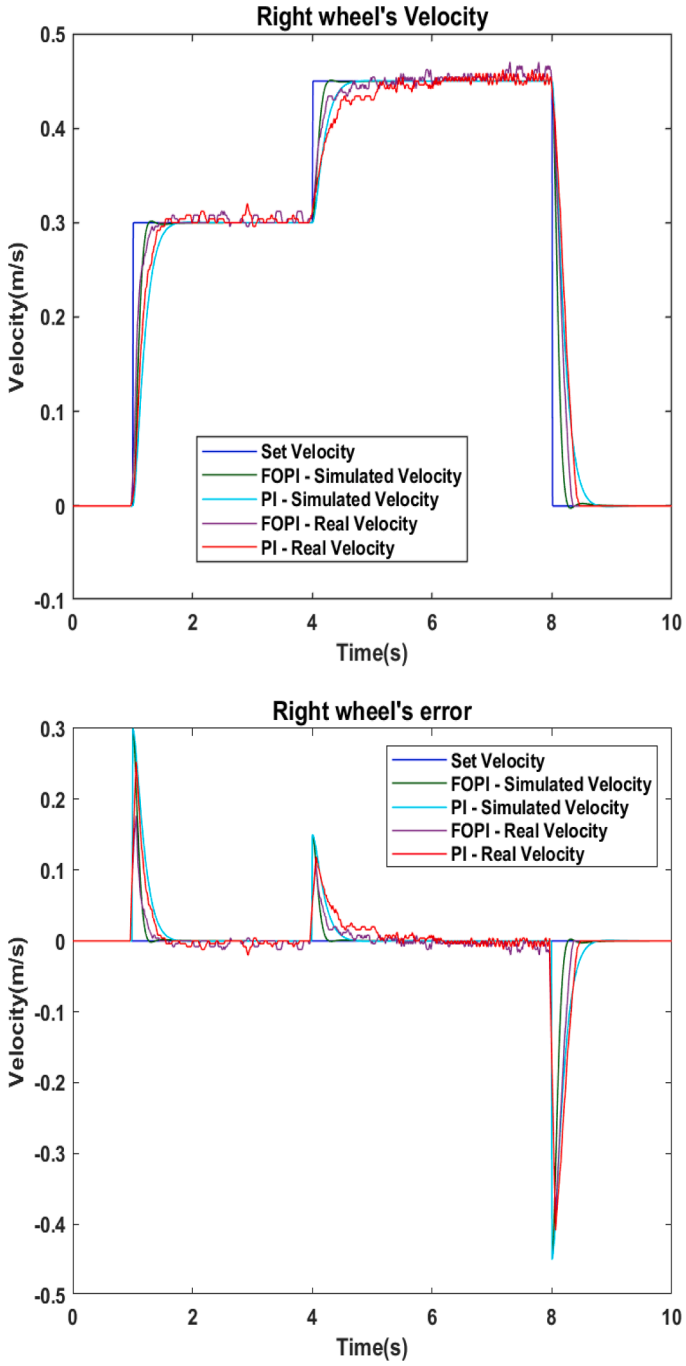


Fig. 11. (a). The compared results of stairs response with GA based FO-PI tuning, PI tuner for simulation and real-time system; (b). The compared errors of stairs response with GA based FO-PI tuning, PI tuner for simulation and real-time system.

with no overshoot in its response. The setting time, the time taken for the system's response to reach and stay within a certain tolerance of the desired value, is shorter for the proposed GA-FOPI controller than for the PI tuner. The GA-FOPI controllers demonstrate fast response times and accurate tracking of the reference input (desired velocity). The outputs of the various GA-FOPI controllers show only slight differences, indicating that the proposed approach consistently provides effective control across different scenarios. In short, the results indicate that the GA-FOPI controller performs better regarding response speed, absence of overshoot, and accurate reference tracking. Fig. 12 shows the 3D map created using D435 and the path generated by the proposed path planning in the small restaurant. The robot can travel to all the ordered tables, recognize the tables, chairs, and humans, and avoid them. From these graphs, it is evident that there is an excellent agreement between simulations and experiments. Fig. 13 reports snapshots of the experiment, where the robots are reported in the robot travelling to serve at different tables and then return to the counter table.

The superior performance of the GA-FOPI controller can be explained by its adaptive tuning capability through genetic optimization. By adjusting the fractional gains, GA-FOPI achieves a better balance between convergence speed and system stability compared with conventional PI and FOPI controllers. This adaptive mechanism helps minimize overshoot, improve steady-state accuracy, and maintain consistent performance even under varying operating conditions. Similarly, the hybrid A*–DWA planner effectively combines global path optimality with local dynamic responsiveness, enabling the robot to navigate smoothly and safely in environments containing static and moving obstacles. The combination of these methods ensures that the robot can plan feasible paths while reacting quickly to environmental changes. The system maintained stable operation under moderate lighting variations and sensor noise. The fusion of RGB-D and IMU data through the EKF helped mitigate inaccuracies from low-cost sensors, contributing to reliable localization and mapping. Despite these advantages, the present work has several limitations. The perception module relied on YOLOv3, which, although sufficient for real-time operation, is less advanced than newer object detectors. The testing environment was limited to a small-scale restaurant setup, and benchmarking against advanced control or planning methods was not included. These limitations are acknowledged as part of the study's current scope. Future improvements will focus on enhancing perception accuracy, expanding testing scenarios to more complex and dynamic environments, and integrating additional control strategies to increase generalizability and robustness.

6. Engineering applications

The proposed integrated framework demonstrates practical applicability in various real-world service-robot scenarios that require reliable, low-cost autonomous navigation. Beyond restaurant environments, the same approach can be extended to indoor delivery robots, hospital logistics systems, and warehouse automation. By relying on affordable sensors and open-source software, the system provides an engineering solution suitable for small and medium enterprises aiming to adopt autonomous robots without high infrastructure costs. The



Fig. 12. 3D service robot's trajectory using the proposed algorithm in the small restaurant.



Fig. 13. Operation of the serived robots: the snapshots of the experiment, where the robots are reported in the robot travelling to serve at different tables and then return the counter table.

modular design also facilitates integration into industrial and educational platforms, offering a scalable foundation for broader applications in intelligent robotics.

7. Conclusion

This study presented an integrated framework for autonomous navigation of a low-cost mobile robot operating in dynamic indoor environments. The system combines odometry, SLAM, hybrid A*–DWA path planning, and a GA-optimized fractional-order PI (FOPI) controller into a unified, real-time architecture implemented on affordable hardware.

The main findings and contributions can be summarized as follows:

- GA-FOPI controller: The proposed controller achieved a rise time of 0.17 s and a settling time of 0.26 s with negligible overshoot (0.01 %), representing >50 % improvement compared with the conventional PI controller.
- Hybrid A*–DWA planner: The integration of global and local planners enabled smooth, collision-free trajectories and reliable navigation in dynamic restaurant-like environments.
- Low-cost platform implementation: All algorithms were deployed on a low-cost embedded system using open-source ROS, demonstrating that high-performance navigation can be achieved without high-end sensors or processors.
- Robustness: The system maintained stable operation under moderate lighting variations and sensor noise. RGB-D and IMU fusion via the EKF effectively mitigated inaccuracies from low-cost sensors.
- System-level novelty: The originality of this work lies in integrating multiple established algorithms into a single cost-effective robotic platform validated through both simulations and real-world experiments.

Despite these advantages, several limitations remain. The odometry accuracy may degrade over long trajectories or in feature-sparse areas, and the current validation was limited to a small-scale restaurant scenario. Future extensions will focus on expanding testing to larger and more complex environments, improving perception with advanced detectors, and evaluating adaptive controllers to further enhance generalizability and robustness.

Overall, the results confirm that the proposed GA-FOPI-based integrated system delivers reliable, real-time, and affordable autonomous navigation performance suitable for practical service-robot applications.

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Ethics declarations

Informed consent was obtained from the individual for the publication of identifiable images in this online open-access publication.

Declaration of AI

During the preparation of this work, the authors used Grammarly to improve readability and language quality. The authors reviewed and edited the content and take full responsibility for the final version.

CRediT authorship contribution statement

Thi-Thoa Mac: Writing – review & editing, Writing – original draft, Project administration, Conceptualization, Funding acquisition, Methodology, Supervision. **Tan-Loc Vu:** Writing – original draft, Visualization, Validation, Software, Data curation. **Hai-Le Bui:** Writing – original draft, Software, Resources, Formal analysis, Data curation. **Xuan-Thuan Nguyen:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Methodology, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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